INCH-POUND

MIL-L-50429A (AR)
Amendment 4
24 May 1990
SUPERSEDING
AMENDMENT 3
8 October 1975

MILITARY SPECIFICATION

LAMINATE, ACETAL AND CLOTH
(FOR USE WITH 81MM AMMUNITION)

This Amendment forms a part of Military Specification MIL-L-50429A (MU), dated 31 October 1972, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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3.2: Add the following to Table I:

	Properties	Requirements	Test Methods
	"Delamination Strength	1/8 lb, min.	4.4.7
	Adhesive Weight acrylic-acetal laminate) (applicable to	0.27 oz/sq yd, max.	4.4.8 or 4.4.9 (alternate method)
*	Adhesive Weight (applicable to acrylic-acetal laminate when used in the M483A1/M509A1 /M718A1/M741A1/M692/M731 or other artillery round)	0.32 oz/sq yd, max.	4.4.8 or 4.4.9 (alternate method)"

3.2: Delete "Basis Weight (applicable to silk-acetal laminate)" and substitute "Gross Weight (silk-acetal laminate)."

3.2: Delete "Basis Weight (applicable to acrylic-acetal laminate) - 3.5 ozs. per sq. yd., max" and substitute "Gross Weight (acrylic-acetal laminate) - 3.4 ozs. per sq. yd., max."

* 3.2: Delete:

Properties	Requirements	Test Methods
"Roll width (Charge A Bag)	2.375 $\pm$ .063 inches	4.4.3
Roll width (Charge B Bag)	2.750 $\pm$ .063 inches	4.4.3"

AMSC N/A FSC 1395

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* Add the following paragraph:

"3.2.1 Width. As specified by the contract or purchase order, when tested as specified in 4.4.3."

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3.6: Delete "Ballistic Comparison Test" where it appears and substitute "Residue Test."

3.7: Delete "Samples" and substitute "First article samples."

4.2.2.1, first sentence: Delete "in accordance with 4.2.2.2."

4.2.2.1, second sentence: Delete "shall consist of one lot" and substitute "shall be selected from one lot."

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4.3: Delete in its entirety and substitute the following:

"4.3 Sampling Plans.

4.3.1 Applicable to Gross Weight (acrylic-acetal laminate) and Adhesive Weight (acrylic-acetal laminate). See paragraph 3.2. A group of five rolls of Charge A Bag width and five rolls of Charge B Bag width shall be randomly selected from each lot. Four samples shall be taken from each Charge A Bag width roll and each Charge B Bag width roll. Samples shall be selected, one from each end and two from the middle of each roll. Compliance with the weight requirements shall be determined after averaging the results of the four samples per each laminate roll. Failure of any roll to comply with the requirements shall cause rejection of the entire lot.

4.3.2 Applicable to all other requirements. See Section 3 REQUIREMENTS. A group of five rolls of Charge A Bag width and five rolls of Charge B Bag width shall be randomly selected from each lot. A sample shall be removed from each roll, tested and inspected for compliance with the requirements of this specification. The sample shall be conditioned as specified in ASTM Designation D618. Failure of any individual sample to conform to the requirements of this specification shall cause rejection of the entire lot."

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4.4.5: Delete in its entirety and substitute the following:

"4.4.5 Residue Test, Special defect (see 6.3). This test shall be conducted at a Government proving ground utilizing 81mm, M374A2 mortar rounds containing propellant increment bags manufactured according to drawings 9233368, 9233369, 9233370 and 9233371. Test rounds shall be assembled with propellant increments made from the first article laminate material sample. Reference rounds shall be assembled with propellant increments selected from an acceptable lot at the time of first article testing. Test rounds and reference rounds shall be assembled utilizing the same metal parts component lots.

All rounds shall be fired at an elevation of 800 mils, from an 81mm mortar tube (3.207 inches max.) for which the round is standard, at a rate of 12 rounds per minute. The firing sequence shall be in accordance with Table II. The mortar tube shall be cooled between the firing of each group. In addition, the tube shall be dry swabbed and elevation checked between the firing of each group. Observance shall be made for hangups. The first article sample will be considered acceptable if no hangups occur. It will then be unnecessary to test the reference rounds. However, if a hangup occurs, it shall be determined if it was caused by residue from the propellant increment bag. Mortar tube elevation shall be checked. A hangup caused by residue will constitute rejection of the first article sample. In this event, the reference rounds will be tested without delay at that charge zone in which the hangup occurred and in the same mortar tube after dry swabbing of the tube. All of the first article test rounds (300 total) shall be tested.

TABLE II

Silk/Acetal Zone	Acrylic/Acetal Zone	Test Rounds	Reference Rounds
1	1	25	25
2	2	25	25
3	6	25	25
2	3	25	25
4	6	25	25
2	4	25	25
5	6	25	25
2	5	25	25
6	6	25	25
7	7	25	25
8	8	25	25
9	9	25	25"

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Add the following paragraph:

"4.4.7 Delamination strength, Major defect. Samples chosen in accordance with paragraph 4.3 shall be one inch wide and twelve inches long. One end of each sample shall be manually delaminated. The delamination strength shall be determined in accordance with ASTM Designation D882, Method A of Standard Methods of Test for Tensile Properties of Thin Plastic Sheeting. The sample shall be mounted in the test machine so that the tabs of the laminated end will be pulled in a direction 180 degrees apart from each other. The rate of separation of the jaws shall be 12 inches per minute. Failure to comply with the requirements as specified in paragraph 3.2 shall be cause for rejecting the laminate lot."

Add the following paragraph:

"4.4.8 Adhesive Weight, Major defect. A 5 inch by 9 inch specimen shall be cut from each laminate sample selected in accordance with paragraph 4.3. The sample shall then be accurately weighed to ± .001 gram. Weighing can be accomplished utilizing a tared 30 ml fritted glass extraction crucible (tall form of coarse porosity). This same crucible (containing the laminate sample) shall then be placed in an extractor (Roweg, Soxhlet or equivalent). Approxinately 80 ml of toluene shall be poured into a 125 ml tared Erlenmeyer extraction flask with a standard taper joint 24/40. Then, this extraction flask shall be fitted to the bottom of the extractor. The top of the extractor shall be attached to a condenser (Allihn type or equivalent). Heat shall be applied to the bottom of the extraction flask containing the toluene. This can be accomplished by means of an electric hot plate (Extraction Heater) which is temperature controlled. Adjust the temperature so that the solvent (toluene and adhesive) drips from the condenser at the rate of two to three drops per second. Extraction shall continue for 6 hours. When extraction is completed, the extraction flask, containing the toluene and extracts, shall be disconnected from the extraction apparatus. Flask should contain approximately 80 ml of toluene solution. Any lesser amount indicates that the apparatus had a leak during extraction. This makes it necessary to conduct this test again. An air hose should be placed in the toluene solution

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through which dry air shall be forced which will evaporate the toluene from the extract. This should continue until the odor of toluene can no longer be detected. The flask should then be placed in a vacuum desiccator over sulphuric acid (or calcium sulfate). The flask and contents shall be weighed every two hours until the change in weight at these intervals does not exceed two milligrams. The gain in weight of the flask is the adhesive weight.

The laminate material remaining in the 30 ml extraction crucible shall be air dried until the order of toluene is no longer detected. The crucible shall then be placed in the vacuum desiccator for four hours minimum. The weight of this material plus the adhesive weight must equal the original laminate material weight within two percent."

Add the following paragraph:

"4.4.9 Adhesive Weight (alternate method), Major defect. A 2 inch X 4 inch specimen shall be cut from each laminated sample selected in accordance with paragraph 4.3. The sample shall be accurately weighed to ± .0002 gram. The sample shall then be immersed in 250 ml of methylene chloride (reagent grade) for one minute. The methylene chloride shall be used for not more than five (5) different laminate samples. The sample shall be withdrawn and separated into the acetal and cloth materials. Both cloth and acetal shall be wiped clean and allowed to dry for five (5) minutes. The immersing, wiping and drying shall be repeated enought time to assure a weight variation no greater than ± .0005 grams. The weight of the adhesive shall then be determined."

Add the following paragraphs:

"6.3 Special defects.

6.3.1 Definition. A special defect is a defect that judgement and experience indicate could, depending on the degree of variance from the design requirement, result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product.

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6.3.2 Disposition. Upon investigation and analysis, of lots exhibiting special defects, by the responsible technical activity, a recommendation of acceptance or rejection shall be made to Director of Quality Assurance, U.S. Army Armament Command. Final acceptance of such lots on waiver shall be made only by the Director of Quality Assurance, U.S. Army Armament Command."

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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